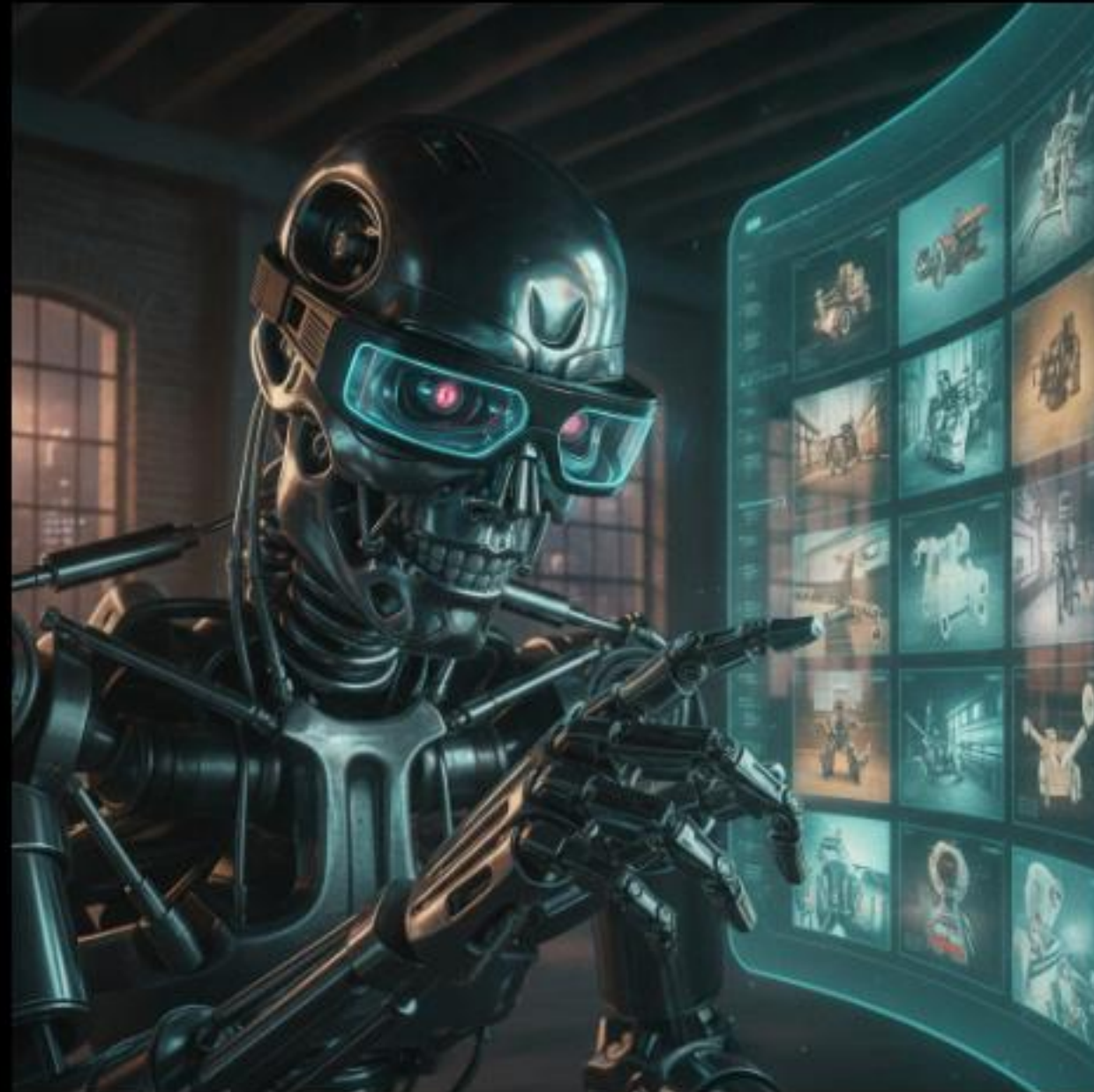


Image Analysis



Tauhid Zaman

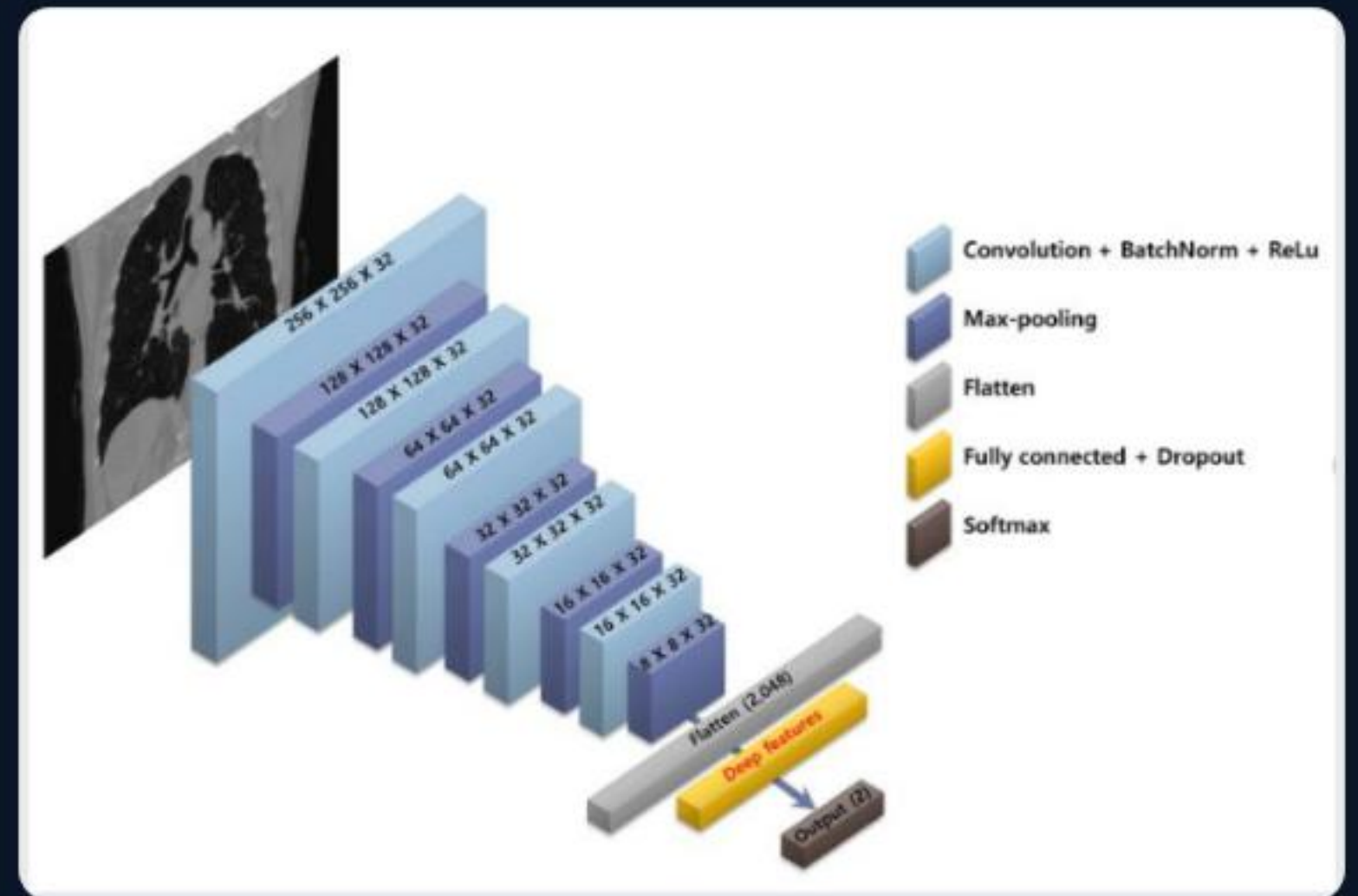
AI Vision

- So far the agent mostly **talks** — text in, text out
- Today we will learn how AI models can **see**: a picture becomes words, data fields, and decisions
- We can use this ability to construct **vision** based tools for our agents



Convolutional Neural Networks ~1989-1998

- Researchers copied the **convolutional** structure of the eye's **visual cortex** to produce convolutional neural networks (CNNs)
- Convolution - an object looks the same (stimulates same neurons) even if it moves it around
- First successful CNN: LeNet (made by Yann LeCun) for reading handwritten digits in 1998



Residual Networks (ResNet) 2015

- Deeper nets should get smarter — but without a trick, the signal **dies** as you stack layers
- **Skip connections:** let information jump over blocks
- ResNet (He et al., 2015) showed that skip connections let you train neural networks with 100+ layers

No skip



With skip



Vision Transformers 2020

- Language models already used **transformers** (attention over tokens)
- Vision Transformer (ViT, Dosovitskiy et al., **2020**): chop the photo into a grid of **patches**
- Treat each patch like a token (just as with text)
- Vision transformers showed equal or better performance to CNNs



Contrastive Language Image Pre-training (CLIP) 2021

- OpenAI, 2021 — trained a transformer to embed images and their captions in the same space
- CLIP model was then able to write captions for new images



Modern Vision AI Models

- Modern AI models can easily understand image content
- You can ask it to describe, extract fields, or answer questions about the image
- Images are billed as **input tokens**
- 1024×1024 image is approximately 1,229 tokens (at high detail)

Model	Cost / 1M input tokens	≈ Cost / image
<code>gpt-5.6-luna</code>	\$0.20	~\$0.00025
<code>gpt-5.6-terra</code>	\$2.00	~\$0.0025
<code>gpt-5.6-sol</code>	\$4.00	~\$0.0049
<code>gpt-6-astra</code>	\$10.00	~\$0.0123

Source: [OpenAI API Pricing](#) · image token rules: [Images & vision](#)



Example — Structured Data From An Image

- Business photo → data fields a **tool** can use
- First define the fields: name, size, condition, shelf location...
- AI will extract the fields from the image based on your prompt

```
{  
  "product": "Aurora Blend",  
  "size": "12 oz",  
  "condition": good  
  "shelf": 2  
}
```



Example — Counting Inventory

- Ice factory wants to count bags of ice produced
- Prompt AI to count bags of ice in image, describing what the bag looks like
- AI can count bags and tell you their location

```
{  
  "product_count": 16,  
  "locations": "10 on cart, 5 on floor, 1 in worker's hands"  
}
```



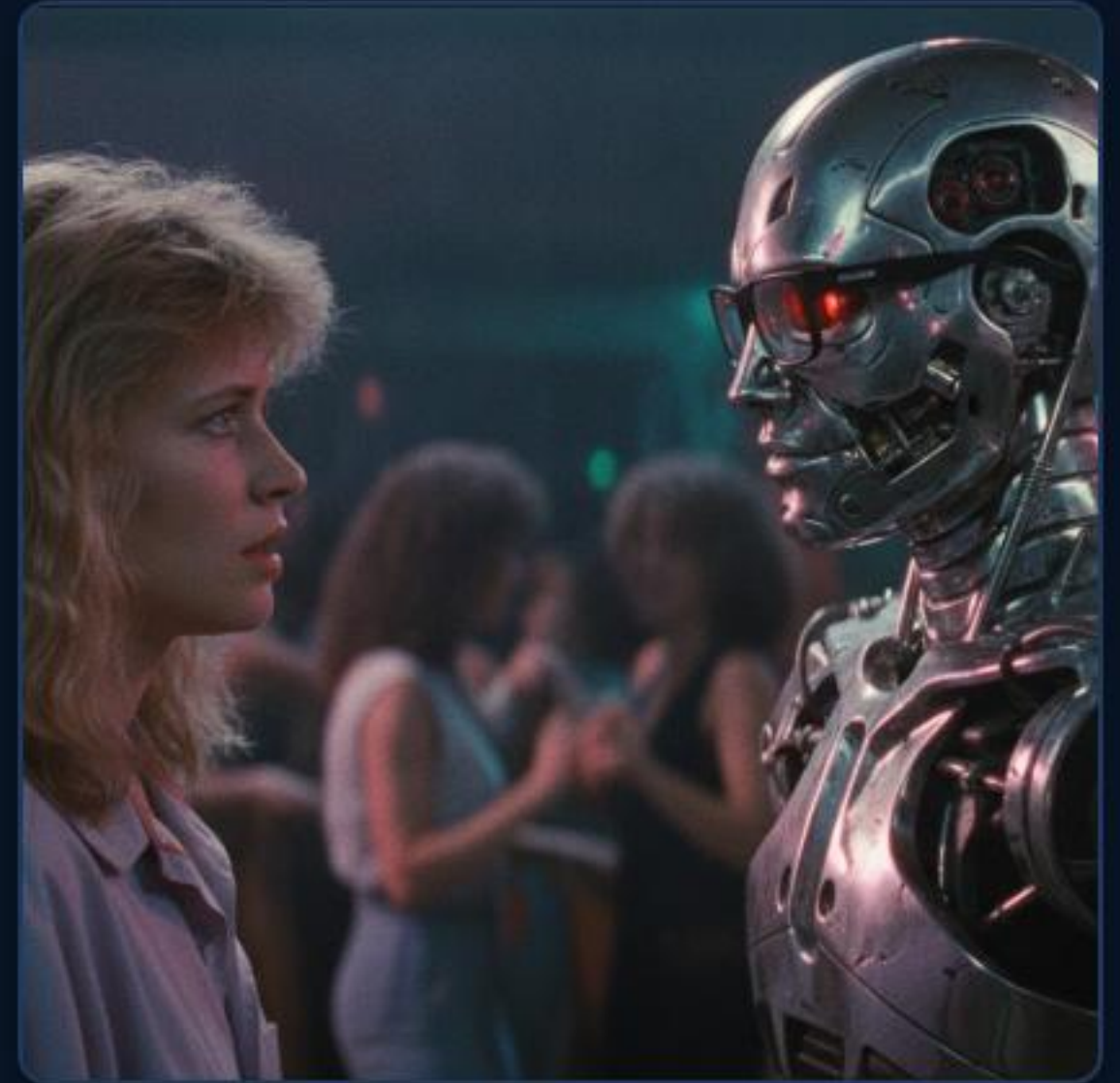
Example — Immersive Browser

- Agent takes **screenshots** of your browser — it sees what you see
- Ask questions about the page: "What does this chart say?" / "Where's the checkout button?"
- Same vision loop: pixels → structured answer → next step
- Useful for help desks, training, and "look at my screen with me"



| Ethics & Privacy Issues

- Vision can be abused: invading privacy, tracking people, recording private data
- Put **safeguards in your agent**: refuse face ID / stalking / secret recording
- Make a list of permitted queries (inventory, products) — refuse “who is that person?” style queries



Vision Is A Tool

- We can build vision tools for our agents
- **PydanticAI** agent flow: vision tool → **structured output** → next tool
- Inventory counting, production monitoring, and more creative applications
- [Vibe coding session](#) →

